Contents

1 Data Analysis Expressions (DAX)

Data Analysis Expression (DAX) is the formula language used in Microsoft Power BI, Power Pivot, and SQL Server Analysis Services (SSAS) Tabular Models. DAX enables analysts and developers to create custom calculations, business metrices, KPIs, and advanced analytical models.
DAX operates on columns, tables and relationships within a semantic model. It combines concepts from Excel formulas, relational databases, and functional programming to provide powerful analytical capabilities.
It allows users to:

- Create calculated columns
- Create measures
- Create calculated tables
- Perform aggregation.
- Perform filtering
- Perform ranking
- Perform statistical analysis
- Perform time intelligence calculations
- Build KPIs

Components of DAX : DAX consists of three major calculation objects:

- **Calculated Columns:** Calculated columns compute values row by row and become permanent columns within the data model. Examples: Profit
- **Measures:** Measures perform calculations dynamically based on filters applied in reports. Examples: Total Sales.
- **Calculated Tables:** Calculated tables generate entirely new tables using DAX expressions. Example: Calendar Table

Characteristics of DAX: DAX is:

- Column-based
- Relationship-aware
- Context-sensitive
- Case-insensitive
- Functional
- Optimized for analytical calculations
- Integrated with VertiPaq Engine

1.1 Create a measure to calculate the total sales amount from the sales table.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX formula:
  - Total Sales = SUM(Sales[SalesAmount])
- Press Enter
- Add the measure to a Card visual

1.2 Create a measure that calculates the total quantity sold.

Answer:

- Create a New Measure
- Enter the DAX
  - Total Quantity = SUM(Sales[Quantity])
- Press Enter
- Create a Card visual and drag: Total Quantity to Card

1.3 Create a calculated column that computes Profit for each sales transaction.

Answer:

- Select the Sales table
- Click New Column
- Enter the DAX expression
  - Profit = Sales[SalesAmount] - Sales[Cost]
- Press Enter

1.4 Create a measure to calculate the average sales amount.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
  - AverageSales=SUM[Sales[SalesAmount]]
- Press Enter
- Add the measure to a Card visual

1.5 Create a measure that counts the total number of sales transaction.

Answer:

- In the Data/Fields pane on the right:
  - Find Sales
  - Click the Sales table
- Create a new measure
- Enter the DAX formula
  - TotalTransactions=COUNTROWS(Sales)
- Create a Card visual
  - Go to Report view
  - Select Card
  - Drag Total Tansactions into the Card

1.6 Create a measure that calculate the minimum sales amount recorded in the sales table.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
  - MinimumSales=MIN(Sales[SalesAmount])
- Press Enter
- Add the measure to a Card visual

1.7 Create a measure that calculate the maximum sales amount.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
  - MaximumSales=MAx(Sales[SalesAmount])
- Press Enter
- Add the measure to a Card visual

1.8 Create a measure to calculate the total profit.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
  - TotalProfit=SUM(Sales[Profit])
- Press Enter
- Add the measure to a Card visual

1.9 Create a measure to calculate the average profit.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
  - AverageProfit=AVERAGE(Sales[Profit])
- Press Enter
- Add the measure to a Card visual

1.10 Create a measure that counts the number of unique customers.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
  - TotalCustomers=DISTINCTCOUNT(Sales[CustomerID])
- Press Enter
- Add the measure to a Card visual

1.11 Create a measure to calculate the total cost of all products sold.

Answer:

- Open the Modeling tab
- Select the Sales tables
- Click New Measure
- Enter the DAX formula
  - TotalCost=SUM(Sales[Cost])
- Press Enter
- Display the measure in a Card visual

1.12 Create a measure that calculate the total discount provided to customers.

Answer:

- Open the Modeling tab
- Select the sales tables
- Click New Measure
- Enter the DAX Formula
  - TotalDiscount=SUM(Sales[Discount])
- Press Enter
- Display the measure in a Card visual

1.13 Create a measure that counts the total number of products sold.

Answer:

- Open the Modeling tab
- Select the sales tables
- Click New Measure
- Enter the DAX Formula
  - ProductsSold=COUNTROWS(Sales)
- Press Enter
- Display the measure in a Card visual

1.14 Create a measure to calculate the average quantity sold per transaction.

Answer:

- Open the Modeling tab
- Select the sales tables
- Click New Measure
- Enter the DAX Formula
  - AverageQuantity=AVERAGE(Sales[Quantity])
- Press Enter
- Display the measure in a Card visual

1.15 Create a measure that calculates the profit margin percentage.

Answer:

- Open the Modeling Tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
  - ProfitMargin%=DIVIDE(SUM(Sales[Profit]),SUM(Sales[SalesAmount]),0)
- Press Enter
- Display the measure in a card visual

1.16 Create a calculated column that classifies each sales as "High Value" or "Norma" based on the Sales Amount.If SalesAmount>=10000 then HighValue Otherwise Normal.

Answer:

- Open Power BI
- Select the sales Table
- Go to the Modeling tab
- Click New Column
- Enter the DAX Formula
  - SalesCategory=IF(Sales[SalesAmount]>=10000,"HighValue","Normal")
- Press Enter
- Display the column in a matrix visual

1.17 Create a calculated columns that combine First Name and Last Name into a Full Names.

Answer:

- Open Power BI
- Select the employees Table
- Go to the Modeling tab
- Click New Column
- Enter the DAX Formula
  - FullName=Employees[FirstName]&" "&Employees[LastName]
- Press Enter
- Display the column in a table visual

1.18 Create a measure to calculate Total Sales only for orders the Sales Amount is greater than Rs.50000.

Answer:

- Open the Modeling Tab
- Select the Employees Table
- Click New Measure
- Enter the DAX Formula
  - HighValueSales=CALCULATE(SUM(Sales[SalesAmount]),FILTER(Sales,Sales[SalesAmount]>50000))
- Press Enter
- Display a measure in a Card visual

1.19 Create a measure that calculate total Sales only for the "Electronic" product category.

Answer:

- Open the Modeling tab
- Select the sales Table
- Click New Measure
- Enter the DAX Formula
  - ElectronicSales=CALCULATE(SUM(Sales[SalesAmount]),Products[Category]="Electronics")
- Press Enter
- Display a measure in a card visual

1.20 Create a measure that returns the average sales amount for each customer.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New measure
- Enter the DAX Formula
  - AverageSalesPerCustomer=DIVIDE(SUM(Sales[SalesAmount]),DISTINCTCOUNT(Sales[CustomerID]),0)
- Press Enter
- Display a measure in a Card visual